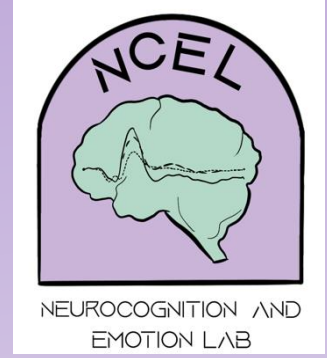


From PTSD to Depression: Longitudinal Shifts in Psychopathology Among 9/11 Responders

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Background

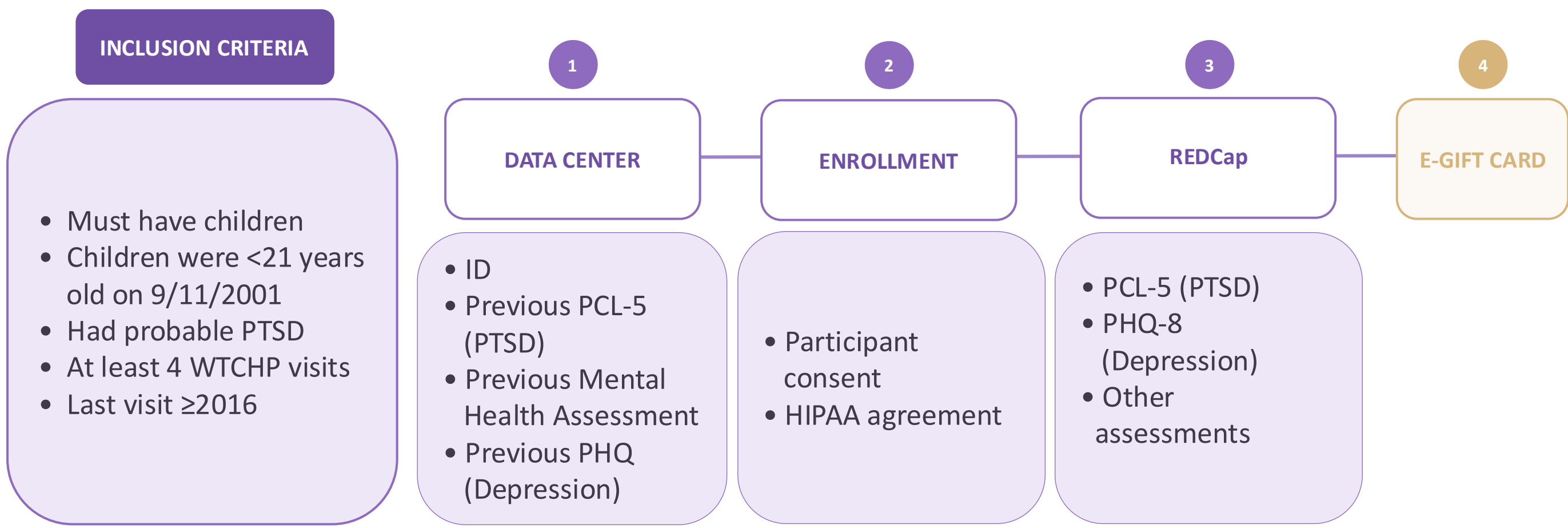
- Exposure to the September 11, 2001 terrorist attacks has been associated with elevated rates of posttraumatic stress disorder (PTSD) among World Trade Center first responders and rescue/recovery workers (WTC-R) (Stellman et al., 2008).
- Early studies primarily emphasized the chronic and persistent nature of PTSD symptoms in this population (Cone et al., 2015).
- Emerging longitudinal evidence suggests that the expression of PTSD symptoms may change over time rather than remain stable, with increasing comorbidity with depression and, in some cases, depression becoming the predominant clinical presentation (Feder et al., 2016).

Aim

This study examines the long-term trajectory of PTSD symptoms in WTC-R, with a specific focus on whether PTSD symptoms attenuate over time and whether depressive symptoms become a more prominent source of impairment.

Study Design

Longitudinal self-report data were used to assess trajectories of PTSD and depressive symptoms in WTC-R across multiple time points.



Analyses

Data analysis was qualitative in nature, and as follows:

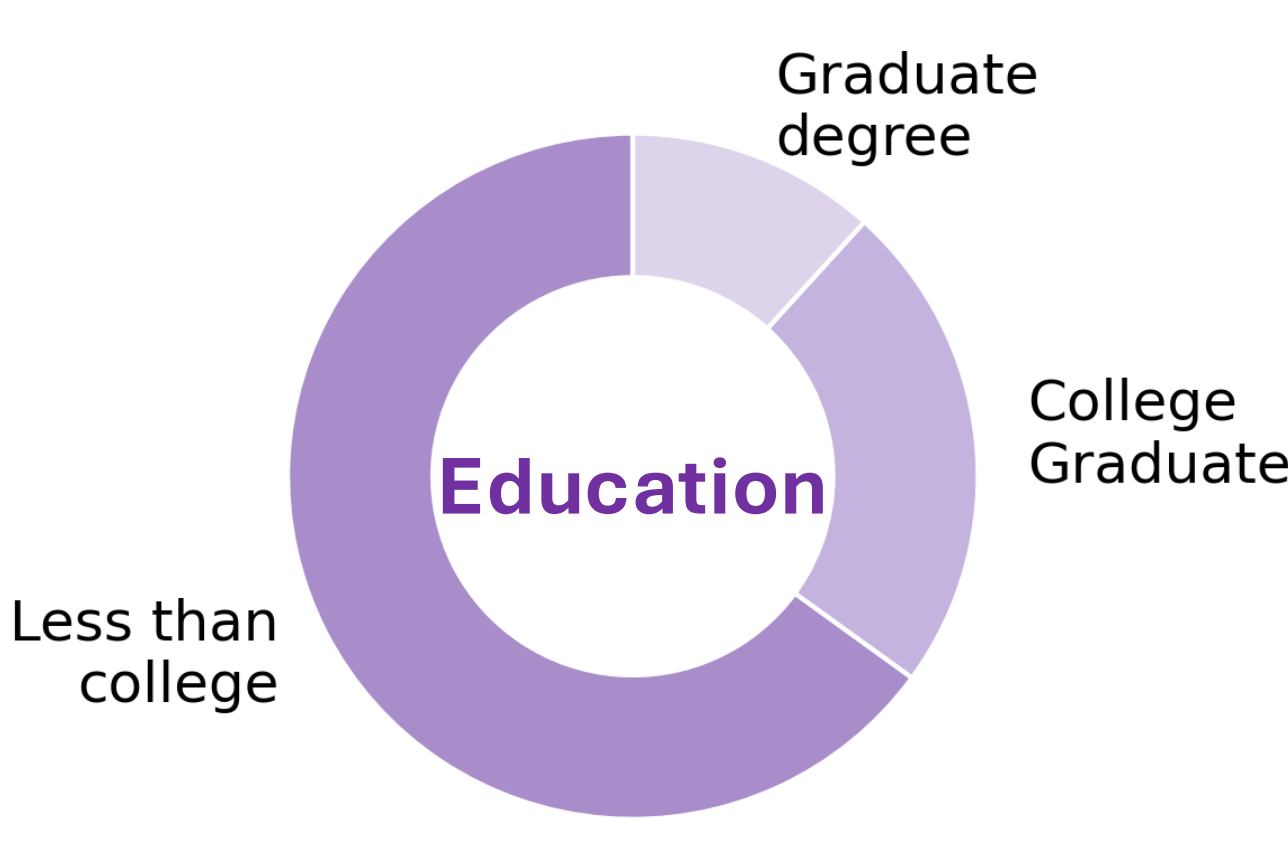
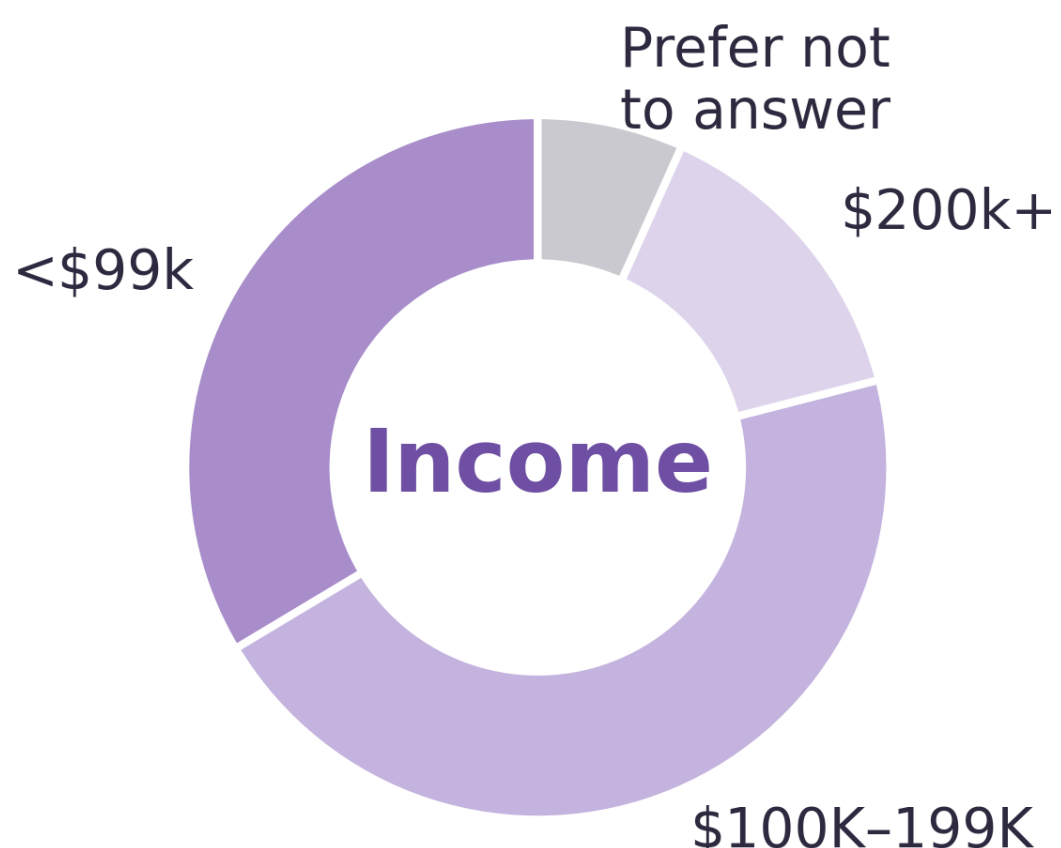
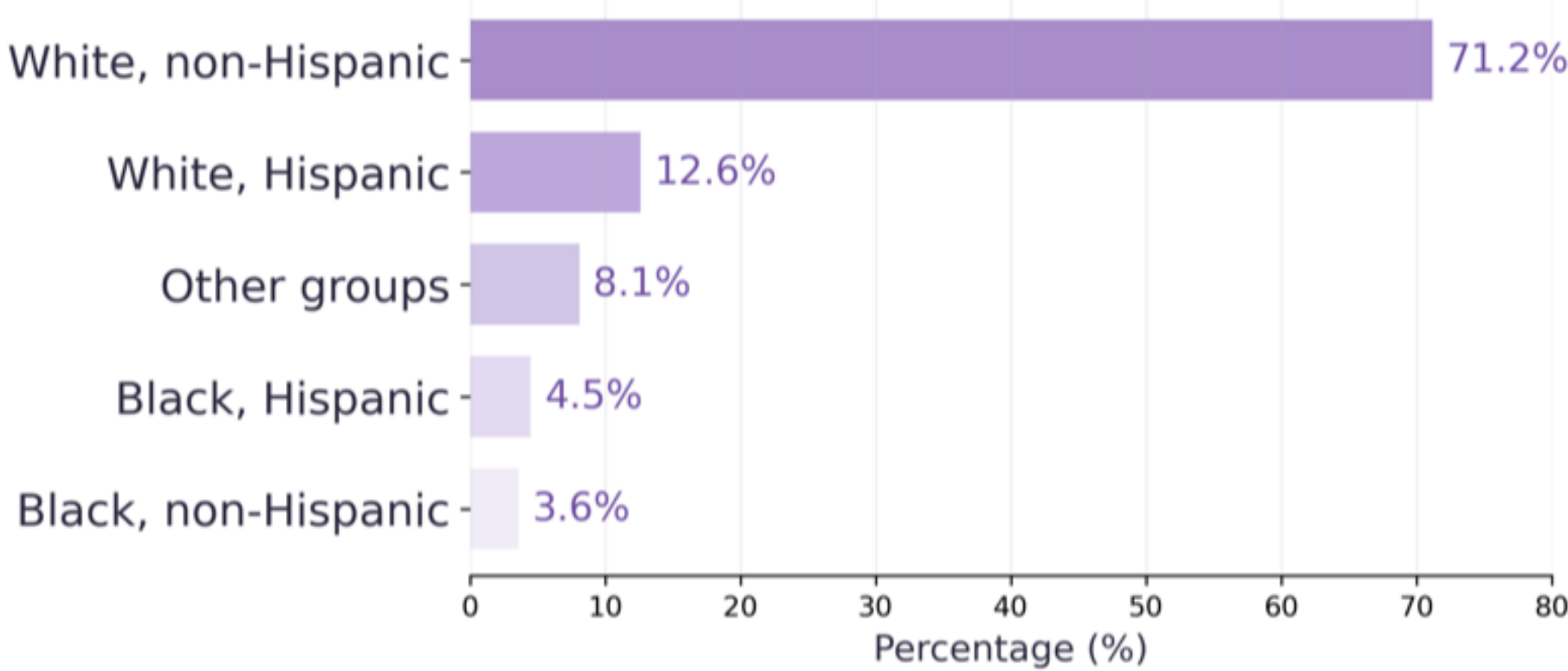
- We reviewed longitudinal data from the GRDC, which included between 4 and 16 visits per participant conducted between 2002 and 2016.
- We integrated additional data from our Intergenerational Transmission of Trauma (ITT) study collected between 2020 and 2021.
- Only participants who completed the PCL assessment in the ITT study were included in the analyses.
- Because the number and timing of visits varied across participants, analyses focused on symptom reporting at three key time points:
 - GRDC Visit 1
 - GRDC Last Visit
 - ITT Study Assessment
- We counted the number of participants reporting PTSD symptoms (PCL) and depressive symptoms (PHQ) at each of these time points.

ITT Sample

Mean age: 60.5 ± 32 years

86.8% male ; 74.8% married ; 18.3% Spanish-speaking

Race & Ethnicity



Results



	Visit 1		Last Visit		ITT	
	PCL	PHQ	PCL	PHQ	PCL	PHQ
N	61	42	66	45	26	50
N comorbidity	33		41		24	
% of comorbidity (relative to PCL)	54%		62%		92%	

- Findings indicate a gradual attenuation of core PTSD symptoms (PCL) over time.
- % of comorbidity with depression (PHQ) increased across assessments.
- For a substantial subset of individuals, depressive symptoms persisted despite reductions in PTSD symptom severity.
- Results suggest a shift in symptom presentation from predominantly PTSD-related symptoms to more chronic depressive symptomatology over time.

Potential Mechanisms

Several mechanisms may underlie this shift:

- As trauma salience diminishes, re-experiencing symptoms may decline, while persistent negative affect and maladaptive cognitions remain, aligning more closely with depressive presentations (Simms et al., 2002).
- Accumulated functional burden (e.g., social withdrawal, occupational difficulties, reduced quality of life), particularly with aging, may increase vulnerability to depressive symptoms (Armenta et al., 2019).
- Prolonged stress exposure and sustained hyperarousal may lead to neurobiological changes associated with depression (Russo & Nestler, 2013).

Conclusions

- These findings highlight the dynamic and evolving nature of trauma-related psychopathology.
- Long-term care for trauma-exposed populations should extend beyond PTSD-focused treatment approaches and incorporate transdiagnostic or staged interventions that address changing symptom profiles, particularly the persistence and progressive increase of depressive symptoms over time.

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